Bojanapalli Kalki Suvarn

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Aditya junior college	Class XII (BIEAP)	2024	98.40%
Aditya(EM) high school	Class X (SSC)	2022	95.10%

SKILLS

Programming Languages: python , C , C++ , bash , SQL

libraries and frameWorks : POSIX , postgresql, numpy , pandas , matplotlib

Tools & platforms: git,github,VSCode , CMake

Core Concept: data structures and algorithms (DSA) , operating systems, DBMS , OOP

PROJECTS

Custom Memory Allocator | Github

- Engineered Free List, Arena, and Pool memory allocation strategies from scratch in C++ to bypass standard malloc overhead and evaluate performance trade-offs in allocator design.
- Achieved low-latency memory management, with the custom Arena Allocator outperforming standard malloc/free by over 50x (221 s vs 12, 267 s) for 100,000 synthetic allocations.
- Implemented custom block splitting and coalescing algorithms via a doubly linked list structure to manage dynamic memory reuse and control heap fragmentation.
- Developed a CMake-configured benchmarking and stress-testing suite to rigorously verify allocation correctness, arena reset behavior, and pool exhaustion under high-frequency workloads.

Technologies / Tools Used : C++, STL, CMake

Mini Unix Shell | Github

- Developed an interactive Unix-like command-line interpreter in C featuring a robust execution loop, custom environment variable mutations (export/unset), and state-aware directory navigation (cd -).
- Designed a quote-aware lexical analyzer and parser capable of processing whitespace-separated tokens, single/double quotes, and unspaced operators (e.g., cat
- Managed multi-process lifecycles via POSIX system calls (fork(), execvp(), waitpid()), enabling safe child process isolation, synchronous foreground execution, and exit state tracking.
- Implemented multi-stage pipeline routing and file descriptor manipulation using pipe() and dup2() to support input (<), output (>), append (»), and arbitrary-depth command chaining.

Technologies / Tools Used : C, Make, POSIX, Linux

Distributed In-Memory Cache Engine | github

- Built a production-style in-memory key-value cache engine with a modular architecture separating storage, eviction, expiration, persistence, and protocol layers.
- Implemented a thread-safe KV store using std::unordered_map and std::shared_mutex to support concurrent read/write operations.
- Designed TTL-based expiration with a background cleanup worker and implemented O(1) LRU eviction using a hash map and doubly linked list.
- Developed snapshot-based persistence for cache recovery and added unit tests using GoogleTest to validate core functionality.
- Architected the engine as the foundation for a future distributed cache supporting consistent hashing, replication, and multi-node deployment.

Technologies/Tools used: C++20 , STL , Multithreading , Shared Mutex , CMake , GoogleTest

Mini Search Engine | Github

- Architected and decoupled a single-file prototype into a modular, production-style C++17 engine spanning independent subsystems for automated web crawling, storage, indexing, and ranking.
- Integrated libcurl to execute live web crawling from seed configuration files, funneling raw source data into a custom HTML parser that builds a Document Object Model (DOM) Tree for text extraction.
- Designed an Inverted Index engine leveraging string-processing algorithms including tokenization, stop-word elimination, and custom word stemming to optimize term-to-document post list dictionary lookups.
- Implemented a vectorized text-ranking pipeline using TF-IDF metric calculations, achieving sub-second terminal query lookups and relevance scoring over compiled document repositories.

Technologies / Tools Used : C++, libcurl, CMake, STL

RELEVANT COURSES

Core : Introduction to programming, Data Structures and Applications, Algorithms, Introduction to computing, Operating Systems, DBMS, Discrete Mathematics for CS

Math : Probability and Random Variables, Statistics , Calculus , Differential equations, Complex variables, Elementary linear algebra